Module End Assignment:

“Analyzing E-Learning Platform Purchases using MySQL”

1. Create the database and schema. Populate the Schema:

```
create database project_db;
use project_db;
create table learners(learner_id int  primary key,
Full_name varchar(40),
Country varchar(50));


create table courses(course_id int primary key,
course_name varchar(40) not null,
category varchar(50) ,
unit_price decimal(10,2));

create table purchases(purchase_id int primary key,
learner_id int,
course_id int,
foreign key (learner_id)
references learners(learner_id),
foreign key(course_id)
references courses(course_id),
Quantity int default 0,
purchase_date date);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:49:28	create database project_db	1 row(s) affected	0.015 sec
2	18:49:28	use project_db	0 row(s) affected	0.000 sec
3	18:52:25	create table learners(learner_id int primary key, Full_name varchar(40), Country varchar(50))	0 row(s) affected	0.047 sec
4	19:00:25	create table courses(course_id int primary key, course_name varchar(40) not null, category varchar(50) , unit_pric...	0 row(s) affected	0.032 sec
5	19:10:54	create table purchases(purchase_id int primary key, foreign key (learner_id) references learners(learner_id), foreign...	Error Code: 1072. Key column 'learner_id' doesn't exist in table	0.000 sec
6	19:15:19	create table purchases(purchase_id int primary key, learner_id int, course_id int, foreign key (learner_id) reference...	0 row(s) affected	0.110 sec

Activate Windows
Go to Settings to activate Windows.

2. *Data Exploration Using Joins*

3. Format currency values to **2 decimal places**.

```
62 •  select course_name ,
63    unit_price  as format_currency
64    from courses;
65
```

Result Grid | Filter Rows: | Export:

course_name	format_currency
Python Programming	25000.00
SQL for Beginners	18000.00
Power BI	30000.00
Excel Basics	12000.00
Machine Learning	45000.00
Python Programming	25000.00
SQL for Beginners	18000.00
Power BI	30000.00
Excel Basics	12000.00
Machine Learning	45000.00

Use **aliases** for column names (e.g., AS total_revenue).

```
• SELECT
      c.course_name AS course_name,
      ROUND(SUM(p.Quantity * c.unit_price), 2) AS
  FROM courses AS c
  JOIN purchases AS p
      ON c.course_id = p.course_id
  GROUP BY c.course_id, c.course_name
  ORDER BY total_revenue DESC;
```

ult Grid | Filter Rows: | Export: | Wrap

course_name	total_revenue
Power BI	90000.00
Python Programming	75000.00
Machine Learning	45000.00
SQL for Beginners	36000.00
Excel Basics	12000.00

Sort results appropriately (e.g., highest total_spent first).

```
80 •    SELECT
81          l.full_name AS learner_name,
82          ROUND(SUM(p.Quantity * c.unit_price), 2) AS total_spent
83      FROM learners AS l
84      JOIN purchases AS p
85          ON l.learner_id = p.learner_id
86      JOIN courses AS c
87          ON p.course_id = c.course_id
88      GROUP BY l.learner_id, l.full_name
89      ORDER BY total_spent DESC;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

learner_name	total_spent
Karthik Raj	78000.00
Rahul Kumar	50000.00
Priya Devi	48000.00
Divya Sri	45000.00
Arun Kumar	37000.00

Combine learner, course, and purchase data.

```
92  select   l.learner_id AS learner_id,
93      l.full_name AS learner_name,
94      c.course_name AS course_name,
95      p.Quantity AS quantity,
96      ROUND(p.Quantity * c.unit_price, 2) AS total_amount
97  FROM learners AS l
98  INNER JOIN purchases AS p
99      ON l.learner_id = p.learner_id
100 INNER JOIN courses AS c
101     ON p.course_id = c.course_id
102 ORDER BY total_amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

learner_id	learner_name	course_name	quantity	total_amount
13	Karthik Raj	Power BI	2	60000.00
15	Rahul Kumar	Python Programming	2	50000.00
14	Divya Sri	Machine Learning	1	45000.00
12	Priya Devi	Power BI	1	30000.00
11	Arun Kumar	Python Programming	1	25000.00
12	Priya Devi	SQL for Beginners	1	18000.00
13	Karthik Raj	SQL for Beginners	1	18000.00
11	Arun Kumar	Excel Basics	1	12000.00

Display each learner’s purchase details (course name, category, quantity, total amount, and purchase date).

```
104 •   select
105     c.course_name,c.category,p.quantity,c.unit_price total_amount,
106     p.purchase_date
107     from learners l
108     inner join purchases p on l.learner_id=p.learner_id
109     inner join courses c on p.course_id=c.course_id;
110
111
112
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_name	category	quantity	total_amount	purchase_date
Python Programming	Programming	1	25000.00	2026-07-01
SQL for Beginners	Database	1	18000.00	2026-09-03
Power BI	Data Analytics	2	30000.00	2026-05-05
Excel Basics	Office Tools	1	12000.00	2026-04-07
Machine Learning	Artificial Intelligence	1	45000.00	2026-08-10
Python Programming	Programming	2	25000.00	2026-09-12
Power BI	Data Analytics	1	30000.00	2026-10-15
SQL for Beginners	Database	1	18000.00	2026-03-18

Display each learner’s total spending (quantity × unit_price) along with their country.

```
114 •   select   l.full_name, l.Country,
115         SUM(p.Quantity * c.unit_price) total_spending
116     FROM learners l, purchases p, courses c
117     WHERE l.learner_id = p.learner_id
118     AND p.course_id = c.course_id
119     GROUP BY l.learner_id, l.full_name, l.Country
120     ORDER BY total_spending DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

full_name	Country	total_spending
Karthik Raj	India	78000.00
Rahul Kumar	Australia	50000.00
Priya Devi	Singapore	48000.00
Divya Sri	Malaysia	45000.00
Arun Kumar	India	37000.00

Find the top 3 most purchased courses based on total quantity sold.

```
124 •   select   c.course_name,   SUM(p.Quantity) total_quantity
125     FROM courses c, purchases p
126     WHERE c.course_id = p.course_id
127     GROUP BY c.course_id, c.course_name
128     ORDER BY total_quantity DESC
129     LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch

course_name	total_quantity
Python Programming	3
Power BI	3
SQL for Beginners	2

. List all learners who have purchased courses from more than one category.

```
132    use project_db;
133    SELECT l.full_name
134    FROM learners l, purchases p, courses c
135    WHERE l.learner_id = p.learner_id
136    AND p.course_id = c.course_id
137    GROUP BY l.learner_id, l.full_name
138    HAVING COUNT(DISTINCT c.category) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name
Arun Kumar
Priya Devi
Karthik Raj

Identify courses that have not been purchased at all.

```
140
141    SELECT c.course_name
142    FROM courses c
143    LEFT JOIN purchases p
144    ON c.course_id = p.course_id
145    WHERE p.course_id IS NULL;
146
147
148
149
```